

Network Situation Awareness Technology based on Extended Berkeley Packet Filter

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Abstract — Network situation awareness technology is an important technology in the field of network security. It can sense and evaluate the real-time status of the network through real-time monitoring, analysis and processing of network data flow. However, the current implementation of network situation awareness technology is not compatible and efficient. To this end, this paper proposes network situation awareness technology based on EBPF. In this paper, we use an optimized Scalable Bloom filter (EBPF) to capture network protocol data and store this data in HBase. Then, we use network situation awareness technology to analyze these data, so as to realize the detection of DOS attacks. Through experiments, this paper verifies the effectiveness and practicability of this method in detecting Denial of service attacks (DDOS), and provides a new idea and method for future network security research.

Keywords: Network Situation Awareness; DDOS detection; EBPF; HBase

I. INTRODUCTION

With the popularity of cloud computing, big data, IoT, network scale and complexity continue to expand, and the importance and necessity of network security situation awareness become more prominent. Network situational awareness technology refers to the comprehensive and accurate monitoring and analysis of network data flow, topology, device status, and discovery and early warning of security incidents through real-time and dynamic collection and analysis, helping network administrators to keep abreast of network operation status and security threats, and take corresponding security measures to ensure the safe and stable operation of the network.

Researchers have explored technology for network situational awareness, proposing techniques such as safety distance (SD) and hierarchical multi-domain network security situational awareness model based on graph database [1,2]. However, these techniques did not fully utilize network information semantics or provide data analysis and extraction means.

To address these limitations, this paper proposes a network situational awareness technology based on EBPF technology, which allows developers to inject small code snippets into kernel functionality without modifying the Linux kernel, extending kernel functionality without source code. EBPF technology is widely used in security, container, and cloud computing fields [3,4,5]. We trust EBPF can address previous

research limitations, for it can extract real-time info from kernel network stack, establish multi-level network situational awareness rules, provide highly structured info, and significantly enhance network situational analysis performance with HBase and MapReduce parallelization technology.

To verify the effectiveness of the method we proposed, the system was deployed in a cluster of ten nodes, and then a DDOS attack was launched on it. Experimental results show that the EBPF-based network Situational awareness technology has the ability to detect X types of DDOS attacks. It is worth noting that although this paper uses DDOS as the laboratory object, the situational awareness technology in this paper is scalable, because all network attacks have the same characteristics. In addition, the design based on the HBASE database in this paper can provide a three-fold analysis acceleration effect at 10 nodes.

In summary, the contributions of this paper are as follows:

- This paper proposes a network situation awareness system based on EBPF, which provides more powerful real-time performance for the network situation awareness system based on EBPF.
- This paper proposes an HBase-based network situational data storage backend, using the capabilities of HBase and MapReduce to accelerate the network situational awareness process.
- This paper proposes to use the EBPF-based network situational awareness system to detect DDOS attacks of 14 network protocols, and obtain a three-fold acceleration under the configuration of a multi-node HBase database.

II. BACKGROUND

A. Network Situational Awareness Technology

Network situational awareness technology is a new type of network security technology that comprehensively and in-depth monitors and analyzes network traffic, device status, and log information to achieve real-time perception, evaluation, and early warning of network security situation. It can grasp the global network security status, discover and prevent various threats and risks in time, and provide scientific and effective support for network security defense[6]. Information

processing uses technologies such as data mining, machine learning, artificial intelligence to analyze and process large amounts of data, identify and extract the characteristics and laws of network attacks, and realize early warning and prevention of network security threats.

B. EBPF technology

EBPF is a virtual machine running in the Linux kernel that can implement functions such as network performance analysis, security monitoring, and container monitoring by loading EBPF programs. EBPF program is a low-overhead, secure, programmable network filter that can run in kernel space, interact with user-space applications, and efficiently obtain and process network data[7,8,9]. EBPF technology allows kernel developers to dynamically inject, modify, and delete network filters without recompiling the kernel or restarting the system, and administrators can easily implement functions such as network traffic analysis, traffic control, and network security detection.

C. HBase and MapReduce

HBase [10] and MapReduce [11] are two commonly used Apache open-source software projects that can be used together for high-performance, scalable distributed data processing and storage. HBase is a distributed, scalable, column-oriented NoSQL database. It is built on the Hadoop Distributed File System (HDFS), which can provide high reliability and high-performance data storage and retrieval in the scenario of massive data storage and high concurrent access. HBase adopts a column-based storage method, which can quickly access data in specified columns, and is suitable for storage of unstructured or semi-structured data. MapReduce is a distributed data processing framework that can process large-scale data sets in parallel. It consists of two steps: Map (mapping) and Reduce (reduction). The Map stage divides the input data into several small blocks, and then processes these blocks in parallel by multiple nodes. The Reduce stage merges the results output by the Map, and finally outputs a summary result. MapReduce can effectively process large-scale data sets, and is suitable for data analysis, processing and transformation.

D. Secure data transmission and trustworthiness judgement

Data security research during network transmission has received more and more attention in the current industry, and many researchers have proposed solutions for data transmission security. In most methods, protecting the security of transmitted data and detecting malicious integrity are usually carried out separately. He et al [12] proposed a secure communication scheme based on quantized synchronization of leader/follower neural networks under an event-triggered strategy. Ding et al. [13] reviewed the latest results of security state estimation and control. And Y. Jiang et al. [14] propose an integrated data-driven framework suitable for control level to handle secure transmission and attack detection at the same time. In this framework, an encryption/decryption method and a trustworthiness judgment method based on security correlation are proposed, and attack sensitivity, introduced delay and design freedom are comprehensively discussed.

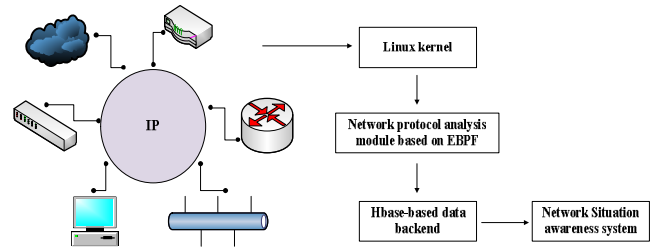


Fig. 1. A framework that combines EBPF and network situational awareness technologies to give a workflow

III. DESIGN AND IMPLEMENTATION

As shown in Figure 1, the framework is mainly divided into the three modules, namely the network protocol data module based on EBPF, the data storage module based on HBase and MapReduce, and the denial-of-service detection system based on network situational awareness technology. First, the technology enters the Linux operating system kernel from the protocol traffic in the network, which is then captured by the network protocol data module based on EBPF, and then stored in the data storage module based on HBase and MapReduce in this paper, and finally performs DDOS attack detection in the network situational awareness system in this paper.

A. Network protocol data module based on EBPF

This article uses the EBPF program to collect some network protocol data needed in this article in the kernel, as shown in Figure 2. These EBPF programs are written in a special language called EBPF. EBPF programs are an instruction set similar to assembly language and can run in the Linux kernel. By adding code to the EBPF program, the kernel can be instructed to perform certain operations, such as capturing network packets, counting system calls, and so on. This article needs to implement the corresponding EBPF program, which can analyze the network protocol header and output the content of the data packet. In order to realize the analysis of the TCP protocol, this article needs to define a structure, which is used to represent each protocol header in the data packet. The structure includes protocol headers such as Ethernet frame header, IP data packet header, and TCP data packet header. By reading the protocol field in the IP protocol header, this paper can judge the protocol type used by the current data packet, and select the corresponding processing logic for data analysis and collection. To load this EBPF program into the kernel, this article uses the load probe function to load the corresponding detection logic defined in the EBPF script. This function compiles the probing logic into a BPF program and loads it into the kernel. Once the EBPF program is loaded into the kernel, it runs when the corresponding event occurs and performs the actions defined in the probe logic, such as capturing network packets and outputting their content.

Using BPF programs and EBPF language to analyze and collect network protocol data can help this paper to better understand the characteristics and behavior of network traffic and detect denial of service attacks. In practical applications, this paper can realize the data analysis and collection of different protocols by customizing the EBPF program and

detection logic according to the needs. This provides strong support for network performance optimization and security assurance.

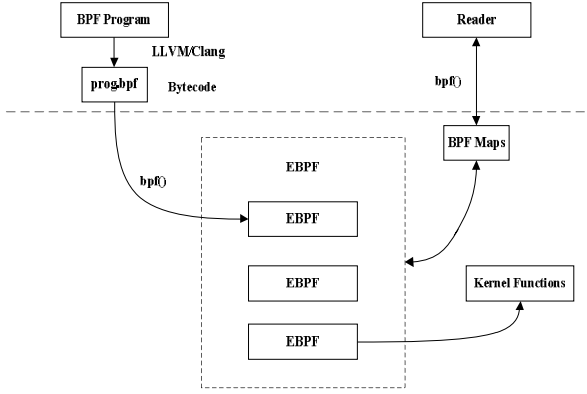


Fig. 2. EBPF

B. Data backend based on HBase

First of all, this article needs to design the structure of the HBase table according to the attributes of the log data. The design of the table structure should take into account the needs of querying and analyzing data. Typically, a row key is used to uniquely identify a row of data, and a column family, column, and value are used to describe attributes of that row's data. For example, information such as timestamps and IP addresses can be used as row keys, column families can be used to store information such as protocol type, source port, and destination port, and columns can be used to store specific log data. The design of the table structure can be optimized according to requirements to improve query efficiency and reduce storage space usage.

Next, this article needs to collect log data for network protocols. Network protocol log data can be captured using EBPF-based tools and saved to a file. When capturing data, this article can specify the type of protocol to be captured and the attributes of the data packet. For example, this paper can capture the data packet of TCP protocol and record its source IP address, destination IP address, source port and destination port and other attributes. The frequency at which data is captured and the amount of data captured can be configured according to requirements.

Then, this article inserts the captured data into the HBase table. Use HBase API or HBase Shell to insert formatted log data into HBase tables. When inserting data, you need to specify the row key, column family, column and value of the data. When inserting data, HBase will automatically partition the data and save it to different Region Servers. This improves data scalability and high availability.

Finally, this article can use HBase API or HBase Shell to query and analyze data stored in HBase tables. When querying data, filters can be used to limit query results to improve query efficiency. For example, this article can use filters to query data within a specified time range or data with a specified source IP address. HBase provides a wealth of filter functions, which can be selected according to requirements. Use some data analysis tools (such as MapReduce) to analyze the data to extract useful

information. For example, this article can analyze the data transfer volume of each IP address or analyze the usage of network protocols in a certain period of time.

C. Network Situational Awareness System

In the network situational awareness system in this paper, advanced techniques and algorithms are used in this paper to parse and analyze the network protocol data in order to gain a deep understanding of the protocol content and structure. By combining the data of network situation awareness, this paper can monitor and analyze network traffic, bandwidth, transmission rate and other information in real time, and detect and defend against DDOS attacks. In terms of DDOS attack detection, this paper analyzes and calculates the statistical data of network protocols based on computer algorithms, so as to establish the detection threshold and heuristic rules of DDOS attacks. At the same time, this paper dynamically adjusts the threshold and rules according to the information of network situation awareness to ensure the accuracy and real-time performance of attack detection.

In addition, for different network protocols, this paper uses advanced analysis technology to further analyze the structured data of the protocol to detect abnormal message segments. These abnormal message segments are often hidden data of DDOS attacks, so this paper can use network situational awareness technology to more accurately discover and defend against DDOS attacks. Finally, this paper monitors the entire network environment in real time and quickly detects DDOS attacks to ensure the security and stability of the network.

IV. EXPERIMENT

In this section, we demonstrate the effectiveness and efficiency of the technology presented in this paper. Firstly, we introduce the experimental setup utilized in this study. Then, we employ simulation tools to replicate diverse DDOS attacks in order to assess the efficacy of the proposed technology. Finally, we validate the efficiency of the EBPF-based Network Situational Awareness Technology.

A. Experiment configuration

Hardware configuration: This article uses 10 Dell R 730 server, R 5218 processing, 32G memory, kernel adopts Linux 5.4.0 version, the specific system is ubuntu 20.04 version.

Software configuration: This article uses LOTC [9] and XOIC [10] to simulate DDOS attacks. LOTC is an open-source network stress testing tool that is also used by some hackers to carry out distributed denial of service (DDOS) attacks. XOIC is also an open-source denial of service attack tool that can run on Windows, Mac and Linux platforms. The tool was developed by a foreign security researcher and is mainly used to test the weaknesses of network applications and systems, but it is also often used by hackers to launch denial-of-service attacks.

B. Validity

This article simulates the DDOS attacks of common network protocols, including IP Flood, SYN Flood and other attacks, as listed in Table 1.

TABLE I. ATTACK TYPES OF DDOS DETECTION BASED ON EPBF-BASED NETWORK SITUATIONAL AWARENESS TECHNOLOGY PROPOSED IN THIS PAPER

<i>serial number</i>	<i>attack type</i>	<i>Test results</i>
1	IP Flood	Detectable
2	SYN Flood	Detectable
3	UDP reflect Flood	Detectable
4	DNS Query Flood	Detectable
5	DNS Reply Flood	Detectable
6	IP Flood	Detectable
7	SYN Flood	Detectable
8	UDP reflect Flood	Detectable
9	DNS Query Flood	Detectable
10	DNS Reply FloodHttps Flood	Detectable
11	Sip Invite Flood	Detectable
12	Sip Register Flood	Detectable
13	Ntp Request Flood	Detectable
14	Connection Flood	Detectable

During the experiment, we carefully recorded attack initiation, cessation, and duration. Using EBPF-based network situational awareness technology, The results demonstrated the system's ability to quickly perceive and categorize DDOS attack traffic in real time, enabling timely defensive measures and effective handling of attack scenarios.

C. Efficient

To validate the effectiveness of our method, we simulat the DDOS attacks in large-scale network environments and use EBPF technology for data collection and analysis, the efficiency and real-time performance of the system. Parameters such as response time and resource usage were recorded to measure the system's performance. The experimental results show that our method is effective in detecting and monitoring various DDOS attacks. The system proficiently identifies DDOS attack traffic through real-time traffic monitoring and attack awareness, and initiates corresponding defense and processing mechanisms. The performance evaluation results once again affirmed the system's outstanding efficiency and real-time performance, even in high-load and large-scale attack environments. It demonstrates rapid response capabilities and can quickly handle large volumes of incoming attack traffic.

V. SUMMARY

The EBPF-based network situational awareness technology we proposed offers a practical and effective solution for cyber security protection. By continuously monitoring and sensing network traffic in real-time, the system promptly identifies and responds to DDOS attacks, guaranteeing network security and stability. Its broad application potential encompasses network operators, enterprises, and Cloud as a Service providers, thereby providing robust cyber security support. Nonetheless, the study also acknowledges certain limitations. Challenges may arise when handling high-density and high-speed traffic, warranting further exploration and optimization. Future

research endeavors should focus on advancing the performance, efficiency, and detection capabilities of the optimized technologies, ultimately strengthening cyber security defenses against emerging threats.

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